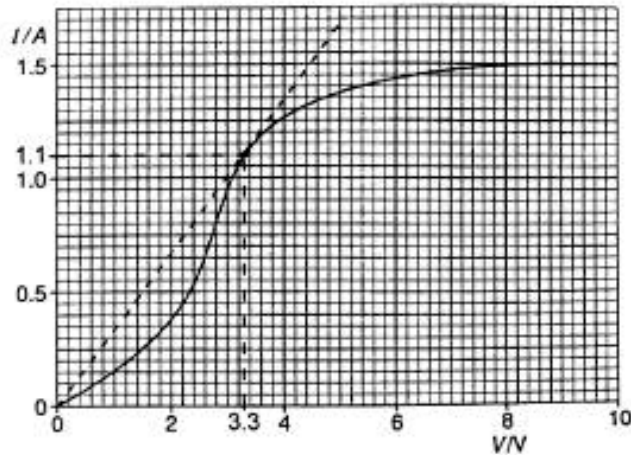


$$\begin{aligned}
 3 \quad (a) \quad (i) \quad R &= \frac{V}{I} \\
 &= \frac{6.0}{1.43} \\
 &= 4.2 \, \Omega
 \end{aligned}$$

- (ii) The minimum value occurs at a point where the ratio of V to I is lowest.
This occurs when $V = 3.3 \, \text{V}$ and $I = 1.1 \, \text{A}$.

$$\text{The resistance } R = \frac{3.3}{1.1} = 3.0 \, \Omega$$



- (b) (i) The p.d. across the component C is the same as that across the $5.0 \, \Omega$ resistor, since both of them are connected in parallel.

$$\begin{aligned}
 V &= IR \\
 &= (0.85)(5.0) \\
 &= 4.25 = 4.3 \, \text{V}
 \end{aligned}$$

- (ii) The total current from the supply is the sum of the current through the $5.0 \, \Omega$ resistor (which is $0.85 \, \text{A}$) and the current through C (from the graph, when the p.d. across C is $4.3 \, \text{V}$, the current through it is $1.3 \, \text{A}$).

$$\begin{aligned}
 \text{Total current} &= 0.85 + 1.3 \\
 &= 2.15 = 2.2 \, \text{A}
 \end{aligned}$$

- (iii) Terminal p.d. $V = E - Ir$

$$\begin{aligned}
 E &= V + Ir \\
 &= 4.25 + (2.15)(0.80) \\
 &= 5.97 \\
 &= 6.0 \, \text{V}
 \end{aligned}$$

- (iv) Energy, $W = Pt = IVt$

$$\begin{aligned}
 &= (1.3)(4.25)(20 \times 60) \\
 &= 6.6 \times 10^3 \, \text{J}
 \end{aligned}$$